Evaluation Validity in Software-Defined Network Intrusion Detection: An Empirical Audit of Controller Observability and Address Generalization on InSDN

Adhyan Jain*
Vellore Institute of Technology
Vellore, India
adhyan.jain2024@vitstudent.ac.in

Atharva Sheersh Pandey*
Vellore Institute of Technology
Vellore, India
atharva.sheersh2024@vitstudent.ac.in

Kshitij Manikshete*
Vellore Institute of Technology
Vellore, India
kshitij.rahul2024@vitstudent.ac.in

Dr. Arivoli A.†
School of Computer Science and Engineering, Vellore Institute of Technology, Vellore, India
arivoli.a@vit.ac.in

***Abstract*—Machine-learning intrusion detection systems (ML-NIDS) for Software-Defined Networks (SDN) are predominantly evaluated on offline flow-record benchmarks computed from end-host captures using randomized train/test splits. We audit the scientific validity of this methodology on InSDN, the public SDN benchmark that distributes the OpenFlow control-channel captures of its ONOS controller, using CICIDS2017 as an external non-SDN reference. Decoding all 44 ONOS control-channel captures (7 381 679 OpenFlow 1.3 messages) reveals that zero flow-table entries and none of the 3 827 FlowMods match IP addresses or transport ports under ONOS's default reactive configuration. Per-flow 5-tuple counters are entirely absent from switch hardware tables; ONOS instead punts IPv4 packets to the controller via an EtherType punt rule present in 44 of 44 captures, delivering 0.68–1.29 PacketIns per target-side packet across 9 of 12 attack scenarios. Bidirectional flows reassembled from PacketIns reproduce packet counts on probing traffic but show only 38% and 45% indicative packet correspondence relative to host-side records on DoS and benign traffic in the examined unsynchronized recordings, with flow duration exhibiting negative rank correlation under flooding ($\rho = -0.27$). While a controller-observable MLP reaches 0.946 balanced accuracy on unseen captures, this reflects controller-side feature separability under capture-grouped evaluation with session-label confounding rather than verified attack detection: InSDN records benign and attack traffic in separate, non-interleaved sessions, where a trivial rule checking for any IP PacketIn scores 0.995 within attack sessions, a host-identity rule learned on training captures reaches 0.734, and classifiers trained on residual traffic after removing attack-matched PacketIns still separate sessions at 0.940 balanced accuracy (residual windows may still contain unmatched attack traffic). Under strict ground-truth labeling, OVS DDoS and OVS brute-force contribute no attack windows, and probing supplies 76% of remaining attack windows. On flow features, randomized 70:30 splits yield 0.9998 (InSDN) and 0.9917 (CICIDS2017) balanced accuracy, whereas source-IP-disjoint evaluation drops to 0.907 ± 0.085 on InSDN (worst split 0.66) and collapses to 0.8417 ± 0.1114 on genuine non-spoofed traffic (Macro-F1 0.6503 ± 0.2005). We resolve the construct validity of address partitioning: 121 938 of InSDN's 122 307 single-flow source IPs belong to the DDoS class, representing ephemeral spoofed addresses generated by an automated flooding campaign from two physical MAC endpoints rather than distinct physical hosts. Combinatorial analysis attributes the 0.66–1.00 bimodal split dispersion to hypergeometric dominant-host allocation ($k = \lceil 0.3M \rceil = 36\,861$, 49.0% theoretical probability of retaining both primary hosts in training, matching 48.0% empirically across 50 seeds, where correlation with test row share is $r = -0.7713$, $p = 5.61 \times 10^{-11}$). As a constructive remedy, square-root frequency attenuation ($w_i \propto 1/\sqrt{N_{\text{host}}}$) acts as a variance-moderating tail hedge (mean paired $\bar{\Delta} = +0.0238 \pm 0.0415$, raising worst-case split from 0.660 to 0.730; Seed 0 from 0.786 to 0.952), whereas pure inverse-frequency ($1/N_h$) collapses to 0.337 under address spoofing. A targeted audit of 12 SDN-IDS papers finds none that reports an address-disjoint split. All findings are bounded to InSDN and ONOS/Ryu. We formulate seven evaluation guidelines and release code and result files for reproducibility.**

***Index Terms*—Software-defined networking, intrusion detection systems, OpenFlow, ONOS, InSDN, control-plane telemetry, evaluation methodology, data leakage, address spoofing, generalization.**

I. Introduction

Software-Defined Networking (SDN) decouples network control logic from forwarding hardware, enabling centralized programmability and dynamic flow management [1], [2]. Motivated by the controller's logically centralized vantage point, an extensive literature proposes machine learning (ML) and deep learning Network Intrusion Detection Systems (NIDS) designed to protect SDN architectures [3], [4], [5], [6], [7].

Despite widespread research activity, a foundational methodological gap separates published experimental evaluations from operational SDN realities. In practice, SDN intrusion detection algorithms are rarely evaluated against active SDN control planes. Instead, researchers predominantly evaluate machine learning models on offline comma-separated value (CSV) benchmarks, such as InSDN [3] and CICIDS2017 [8].

*Adhyan Jain, Atharva Sheersh Pandey, and Kshitij Manikshete contributed equally to this work.
†Dr. Arivoli A. is the corresponding author (e-mail: arivoli.a@vit.ac.in).

These benchmark tables consist of bidirectional 5-tuple flow summaries computed by feature extractors like CICFlowMeter [9] operating directly on full-packet captures obtained from network taps at end hosts. Standard pipelines train high-capacity classifiers on 78+ statistical features—including inter-arrival time moments, bidirectional packet length distributions, and cumulative TCP flag counts—and consistently report detection accuracies exceeding 99.5% [6], [10], [7], [4], [5].

This prevalent methodology suffers from two fundamental threats to scientific validity, illustrated conceptually in Fig. 1:

1) **The Controller Observability Disconnect:** An OpenFlow controller does not have native access to packet-level timing moments or fine-grained per-flow payload metrics. Under standard OpenFlow 1.3 reactive forwarding [11], the controller receives `PacketIn` messages for initial flow misses, issues `FlowMod` entries, and periodically polls aggregate flow statistics counters. Whether the controller's installed flow table entries actually capture fine-grained 5-tuple counters depends on the controller's forwarding application. If the controller installs wildcard or MAC-pair rules, granular 5-tuple statistical features cannot be constructed at the controller without either altering flow-table granularity (risking TCAM exhaustion) or streaming full payloads (risking control-channel saturation).
2) **The Evaluation-Validity and Address Leakage Disconnect:** In security machine learning, randomized train/test splits are known to suffer from severe data leakage when applied to benchmark datasets with repetitive traffic or concentrated host topologies [12], [13], [14], [15], [16]. When flows from the same IP address or session appear in both training and test partitions, models learn spurious address-specific shortcut representations rather than generalizable attack signatures [17], [18]. In SDN topologies with highly concentrated host structures, reported near-perfect accuracies may simply reflect the model's ability to recognize previously seen addresses rather than novel attackers.

While prior studies have highlighted feature calculation discrepancies [19], [20], [21], [22] or identified data smells in general non-SDN datasets [15], [16], the specific manifestation of these issues within SDN control-plane environments remains largely unquantified. Crucially, the InSDN dataset release included not only offline flow CSVs but also the raw PCAP files of the ONOS controller's OpenFlow 1.3 control channel across all evaluated scenarios [3]. This provides a unique, empirical opportunity to examine what an SDN controller actually observes during benchmark execution and to systematically audit evaluation validity.

In this work, we audit controller observability, ground-truth alignment, and evaluation validity for SDN intrusion detection on InSDN, with CICIDS2017 as an external non-SDN reference and comparisons against recent SDN telemetry benchmarks [23], [24], [25], [26]. We make six primary contributions, each an empirical measurement and methodological critique rather than a new detector:

1) **Control-channel characterization (§IV).** We decode all 44 ONOS OpenFlow 1.3 captures shipped with InSDN (7 381 679 messages). No flow-statistics entry (24 437 MAC-pair and 80 464 `eth_type` observations) and none of the 3 827 FlowMods matches IP addresses or transport ports, so the labelled 5-tuple flows are not available as switch counters under this configuration. An IPv4-to-controller punt rule is present in all 44 captures, and the controller receives 0.68–1.29 PacketIns per target-side packet in 9 of 12 attack scenarios.
2) **What PacketIns can reconstruct (§IV).** A custom software flow reassembler with CICFlowMeter-inspired expiration rules, run on one capture per scenario, recovers packet counts and durations in some scenarios and not in others, and its output is incomplete relative to host-side records (showing 38% indicative packet correspondence on DoS and negative duration rank correlation $\rho = -0.27$ in the examined unsynchronized recordings, Table I).
3) **A validity analysis of controller-side detection (§V–§VII).** We show how label construction shapes the evaluated windows (Table II) and compare a controller-side MLP with an "any IP PacketIn" rule, host-identity rules learned on training captures, and a session probe. The 0.946 score cannot be taken as evidence of attack detection, because the benchmark confounds session identity with the label (trivial IP rule scores 0.995 within attack sessions; session probe scores 0.940 on residual traffic after removing attack-matched PacketIns; residual windows may still contain unmatched attack traffic).
4) **Multi-protocol evaluation and spoofing disentanglement (§VIII).** On InSDN and CICIDS2017, random flow splits give ≥ 0.9917 balanced accuracy, whereas source-IP-disjoint splits give 0.907 ± 0.085 and 0.641 ± 0.056. We disentangle address spoofing: 121 938 of InSDN's 122 307 single-flow IPs are spoofed DDoS flows originating from two physical MAC endpoints. On genuine non-spoofed traffic, source-IP-disjoint performance collapses to 0.8417 ± 0.1114 (Macro-F1 0.6503 ± 0.2005). Destination-IP-disjoint splits yield 0.8727 ± 0.0881, and pair-disjoint splits yield 0.9891 ± 0.0084 (Table XIV).
5) **Exact hypergeometric derivation and variance-moderated weighting (§VIII).** We derive the exact hypergeometric formula governing group partitioning under scikit-learn's ceiling rule ($k = \lceil 0.3M \rceil = 36\,861$), explaining the bimodal split dispersion across seeds (49.0% theoretical probability of retaining both primary hosts in training, matching 48.0% empirically across 50 seeds, Table XV, where $r = -0.7713$, $p = 5.61 \times 10^{-11}$). Furthermore, we analyze sample weighting mechanics: square-root frequency attenuation ($w_i \propto 1/\sqrt{N_{\text{host}}}$) acts as a variance-moderating tail hedge ($\bar{\Delta} = +0.0238 \pm 0.0415$, raising the worst split from 0.660 to 0.730; Seed 0 from 0.786 to 0.952, Table XVI), providing a viable alternative to pure inverse-frequency ($1/N_h$), which collapses to 0.337 under extreme address spoofing.
6) **Forwarding mechanism, literature audit, and recent benchmarks (§IX, §X).** A Mininet/Ryu case study demonstrates how forwarding granularity (MAC-pair vs. 5-tuple) and idle timeouts shift controller telemetry vol-

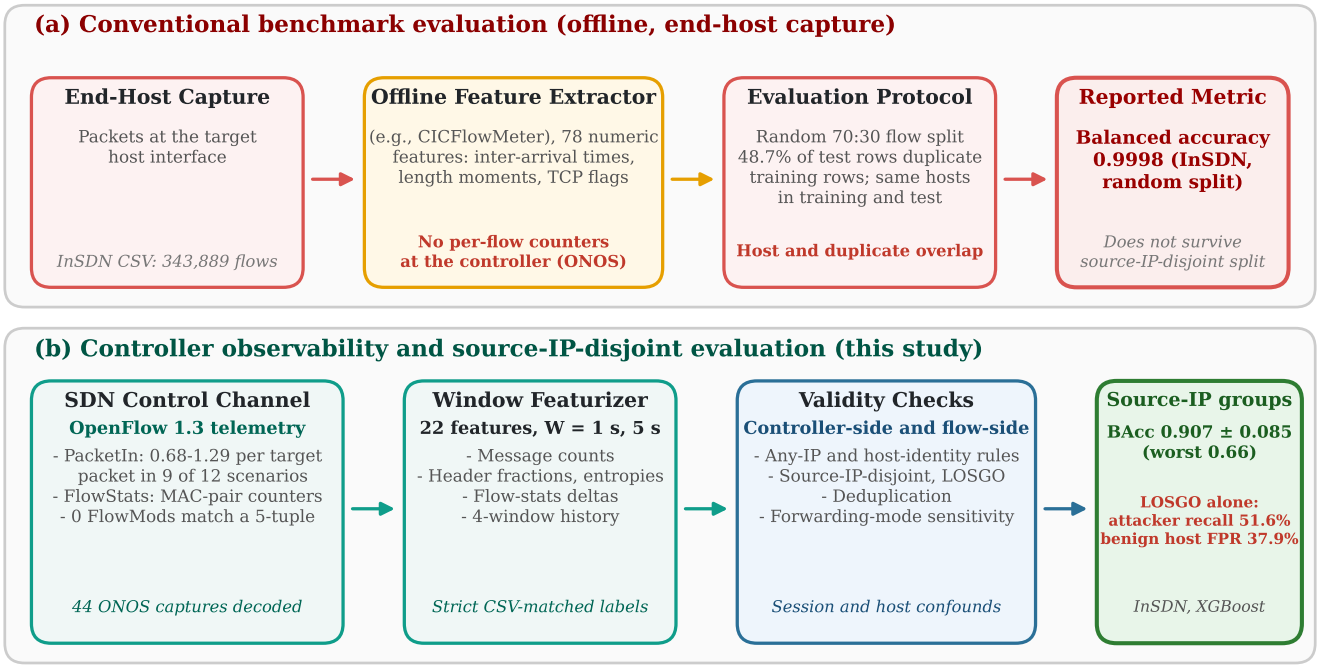


Fig. 1. Overview of the audit. (a) Conventional evaluation computes flow features offline from end-host captures and splits flows at random, with address overlap, 48.7% duplicate test rows and 0.9998 balanced accuracy on InSDN. (b) This study decodes the controller's OpenFlow 1.3 control channel (no flow-table entry or FlowMod matches a 5-tuple under ONOS's configuration), evaluates a controller-side detector against identity baselines, and repeats the flow-feature evaluation under source-IP-disjoint partitioning, where balanced accuracy on InSDN is lower and far more variable.

ume by over three orders of magnitude. A targeted audit of 12 SDN-IDS papers reveals recurring evaluation flaws, four feature recipes approximate published pipelines, and we compare findings against 2024–2026 SDN datasets.

Scope and non-claims. We do not claim a deployable detector, evidence about controllers other than ONOS and Ryu, or causal identification of the host effect. Section XII details the threats to validity.

The remainder of this paper is organized as follows. Section II reviews background and related work. Section III formalizes the threat model, telemetry semantics, and problem formulation. Section IV examines controller observability in InSDN. Section V details the controller-observable detection pipeline, and Section VI presents detection results. Section VII analyzes benchmark session confounds. Section VIII presents the evaluation-validity audit on InSDN and CICIDS2017. Section IX details the Mininet/Ryu forwarding-application case study. Section X presents the targeted literature audit and feature-recipe evaluations. Section XI outlines methodological guidelines and an evaluation checklist. Section XII discusses limitations, and Section XIII concludes the paper.

II. Background and Related Work

A. SDN Control-Plane Mechanics and Telemetry

In an OpenFlow 1.3 network [11], switches maintain flow tables containing match rules, priorities, timeouts, and actions. When a packet arrives at an OpenFlow switch without matching an existing flow entry, it triggers a table-miss event. Under reactive forwarding configurations, the switch encapsulates the packet header (or the entire frame) into an asynchronous `OFPT_PACKET_IN` message and transmits it across the secure control channel to the controller. The controller's forwarding logic processes the packet headers, determines the forwarding path, and responds with an `OFPT_PACKET_OUT` message to forward the buffered frame. Concurrently, the controller may issue an `OFPT_FLOW_MOD` message to install a matching flow entry in the switch's flow table.

Subsequent packets belonging to the same flow match the installed entry directly in hardware or kernel space (such as Open vSwitch datapath), bypassing the controller entirely. The controller can monitor data-plane activity by issuing multipart `OFPMP_FLOW` requests to poll flow statistics. Switches respond with `OFPMP_FLOW` replies containing packet and byte counters for currently active flow entries. When an entry exceeds its idle or hard timeout, the switch evicts it and may send an `OFPT_FLOW_REMOVED` notification.

Crucially, the granularity of information visible to the controller via flow statistics is entirely dictated by the match fields specified in preceding FlowMods by the forwarding application. If an application installs coarse MAC-pair entries (`in_port, eth_src, eth_dst`), all transport connections and IP flows between those two endpoints aggregate into a single counter. Conversely, if an application installs granular 5-tuple entries, flow table occupancy increases dramatically, exposing switches to flow-table saturation attacks.

B. Pitfalls in Machine Learning Security and Network Datasets

The vulnerability of supervised machine learning to subtle experimental artifacts in computer security was seminalized by Sommer and Paxson [12]. More recently, Arp et al. [13] systematized ten common pitfalls in security ML, emphasizing sampling bias, data snooping, and spurious correlations.

In network intrusion detection specifically, benchmark datasets have faced substantial scrutiny. Engelen et al. [14] discovered critical flow construction, labeling, and TCP sequence errors in CICIDS2017, explicitly warning that non-statistical identifier columns (such as Flow IDs, IP addresses, and timestamps) allow classifiers to memorize host identities rather than attack patterns. Lanvin et al. [27] and Liu et al. [28] demonstrated that correcting label and orchestration bugs significantly shifts published model rankings. Flood et al. [15] surveyed seven benchmark datasets (excluding SDN datasets) and introduced the concept of "bad design smells," proving that standard randomized train/test splits result in widespread cross-split leakage due to synthetic traffic homogeneity. Roy et al. [17] demonstrated analogous shortcut learning dynamics on IoT intrusion benchmarks, while Berhili et al. [16] systematically proved across three non-SDN datasets that protocol partitioning choices inflate detection accuracy. Lew et al. [18] highlighted the failure of closed-world evaluations to reflect operational deployment readiness.

C. Intrusion Detection in SDN Environments and Recent Datasets

Early SDN security research focused on leveraging control-plane events to mitigate volumetric denial-of-service attacks. Braga et al. [29] proposed lightweight DDoS detection using 6-tuple flow statistics polled from NOX controllers. Mousavi and St-Hilaire [30] utilized PacketIn entropy variations for early DDoS detection. To protect the SDN control plane against saturation attacks, architectures such as AVANT-GUARD [31], FloodGuard [32], and SPHINX [33] introduced connection migration and flow-table constraint validation.

With the release of benchmark datasets such as InSDN [3], research shifted heavily toward applying advanced deep learning models to flow-level CSV features. Ataa et al. [4] evaluated CNN-LSTM and Transformer architectures on InSDN, reporting accuracies up to 99.02%. Saykat et al. [6] proposed OptiXGB-IDS, combining WGAN-GP oversampling with XGBoost to achieve 99.87% accuracy on InSDN. Mehmood et al. [7] reported 99.98% accuracy using optimizer-equipped CNN-MLPs. Ashikuzzaman et al. [5] introduced SDNGuard-Stack, reporting 99.98% accuracy using stacked ensemble learning. In broader dataset surveys, Ashraf and Masoodi [26] cataloged 124 communication intrusion datasets, highlighting structural evaluation gaps. Recent SDN dataset efforts include SDNFlow [25], which collected OpenFlow 1.5 switch statistics, LAN-SDN-NIDS [23], which fused 11 control-plane metrics into flow CSVs, and ASEADOS-SDN-IoT [24], which captured multi-class IoT attacks in an ONOS testbed. However, as we document in Section X, published evaluations evaluate models exclusively on offline flow records and rely on randomized flow-level splits, leaving their operational validity unexamined.

III. Threat Model and Problem Formulation

A. System Architecture and Telemetry Semantics

Consider an SDN architecture comprising a set of OpenFlow switches $\mathcal{S}$ controlled by a logically centralized controller $\mathcal{C}$ over OpenFlow 1.3 control channels. The data plane carries traffic between end hosts $\mathcal{H} = \mathcal{H}_{\text{benign}} \cup \mathcal{H}_{\text{attack}}$.

The control channel between switch $s \in \mathcal{S}$ and controller $\mathcal{C}$ carries a discrete sequence of timestamped OpenFlow messages:

$$\mathcal{M} = \{(t_i, \tau_i, m_i)\}_{i=1}^{N}, \tag{1}$$

where $t_i \in \mathbb{R}^+$ denotes arrival timestamp, $\tau_i \in \{\text{PacketIn}, \text{PacketOut}, \text{FlowMod}, \text{FlowRemoved}, \text{StatsReply}\}$, and m_i represents message payload. When $\tau_i = \text{PacketIn}$, m_i encapsulates frame header $h_i = (\text{mac}_{\text{src}}, \text{mac}_{\text{dst}}, \text{eth_type}, \text{ip}_{\text{src}}, \text{ip}_{\text{dst}}, p_{\text{src}}, p_{\text{dst}}, \text{proto})$. When $\tau_i = \text{StatsReply}$, m_i conveys flow table records $e_{i,j} = (\text{match}_{i,j}, \text{pkts}_{i,j}, \text{bytes}_{i,j}, \text{dur}_{i,j})$.

B. Problem Formulation: Streaming Window Detection

Rather than assuming omniscient offline flow records, a controller-side detector operates on discrete time windows of duration W. For each window k spanning $[kW, (k+1)W)$, the detector computes a feature vector:

$$\mathbf{x}_k = \phi(\mathcal{M} \cap [kW, (k+1)W)) \in \mathbb{R}^d, \tag{2}$$

where $\phi(\cdot)$ extracts observable control-plane metrics. Given a history of H preceding windows, the model infers a binary or multi-class label $\hat{y}_k \in \{0, 1\}$.

C. Data Partitioning Formulations

To evaluate model generalization, we formalize five distinct partitioning strategies over a dataset $\mathcal{D} = \{(\mathbf{x}_n, y_n, \mathbf{a}_n)\}_{n=1}^{|\mathcal{D}|}$, where $\mathbf{a}_n = (\text{src_ip}_n, \text{dst_ip}_n)$ denotes the address tuple:

1) **Protocol A (Random Flow Split):** Partitions $\mathcal{D}$ into disjoint subsets $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$ via uniform random sampling. Crucially, the host and address spaces overlap heavily: $\text{Addrs}(\mathcal{D}_{\text{train}}) \cap \text{Addrs}(\mathcal{D}_{\text{test}}) \approx \text{Addrs}(\mathcal{D})$.
2) **Protocol B (Source-IP-Disjoint Split):** Partitions the source IP space $\mathcal{A}_{\text{src}}$ into disjoint subsets $\mathcal{A}_{\text{src, train}}$ and $\mathcal{A}_{\text{src, test}}$ such that $\mathcal{A}_{\text{src, train}} \cap \mathcal{A}_{\text{src, test}} = \emptyset$. Records whose source IP belongs to $\mathcal{A}_{\text{src, train}}$ form $\mathcal{D}_{\text{train}}$, and records in $\mathcal{A}_{\text{src, test}}$ form $\mathcal{D}_{\text{test}}$.
3) **Protocol $\mathbf{B}_{\text{dst}}$ (Destination-IP-Disjoint Split):** Partitions the destination IP space $\mathcal{A}_{\text{dst}}$ such that $\mathcal{A}_{\text{dst, train}} \cap \mathcal{A}_{\text{dst, test}} = \emptyset$.
4) **Protocol $\mathbf{B}_{\text{pair}}$ (Pair-Disjoint Split):** Partitions the directed communication pair space $(\text{src_ip}, \text{dst_ip})$ such that no pair appears in both training and test sets.
5) **Protocol $\mathbf{B}_{\text{nospoof}}$ (Spoofed-Separated Split):** Isolates single-packet spoofed addresses from legitimate recurring hosts, evaluating source-IP-disjoint generalization strictly on non-spoofed traffic.
6) **Leave-One-Source-IP-Group-Out (LOSGO):** For each dominant address group $a^* \in \mathcal{A}_{\text{dom}}$, test set is $\mathcal{D}_{\text{test}} = \{(\mathbf{x}_n, y_n) \mid \text{src_ip}_n = a^*\}$, and training set is all remaining records $\mathcal{D}_{\text{train}} = \{(\mathbf{x}_n, y_n) \mid \text{src_ip}_n \neq a^*\}$.

7) **Time-Ordered Split:** For a temporal dataset, records are ordered by timestamp $t_1 \leq t_2 \leq \cdots \leq t_N$, and a cutoff T_{split} defines $\mathcal{D}_{\text{train}} = \{(\mathbf{x}_n, y_n) \mid t_n < T_{\text{split}}\}$ and $\mathcal{D}_{\text{test}} = \{(\mathbf{x}_n, y_n) \mid t_n \geq T_{\text{split}}\}$.

D. Threat Model and Scope

Defender Capabilities: The defender deploys an intrusion detection module co-located with or tapping the SDN controller interface. The detector inspects solely OpenFlow control-plane traffic $\mathcal{M}$. It possesses no packet taps on data-plane switch ports and cannot read host-side sockets. During any given evaluation epoch, the forwarding application running on the controller is assumed fixed and known, though operational changes or policy adjustments require recalibrating detector thresholds as discussed in Section IX.

Attacker Model: The attacker controls one or more compromised end hosts in the network. The attacker initiates network attacks including volumetric flooding (DoS/DDoS), port scanning, service probing, brute-force authentication, web application exploitation, and botnet command-and-control communication. The attacker may randomize or spoof IP source addresses during volumetric flooding. The attacker does not compromise the controller kernel or directly forge control-channel TLS certificates.

Out of Scope: Real-time inline mitigation latency, hardware switch TCAM exhaustion, and adversarial evasion of control-plane telemetry features are outside the scope of this study.

IV. What the Controller Sees in InSDN

A. Dataset Inventory and Capture Reassembly

InSDN [3] was generated in a virtualized testbed using Mininet and Open vSwitch (OVS) connected to an ONOS controller [34]. Traffic targets a Metasploitable-2 virtual machine or the OVS switch itself. In addition to releasing CSV flow files, the authors published raw PCAP archives recorded at the target hosts and at the ONOS controller's OpenFlow interface.

We fetched all 44 ONOS controller packet captures directly from the institutional repository, verifying each archive against published SHA-256 digests. As summarized in Table III, the 44 captures encompass 11 benign sessions and 33 attack sessions spanning seven distinct attack classes.

B. Match Field Granularity and FlowMod Analysis

To determine what flow entries were active at the switches, we parsed and decoded every OpenFlow 1.3 multipart flow-statistics reply across all 44 controller captures. Over the entire dataset, only two distinct match structures appear:

1) `eth_type`: Default punting rules installed by ONOS (80 464 entry observations). Across all 44 captures, our full flow-table decode shows that these 80 464 entries partition equally across exactly four EtherTypes (20 116 observations each): IPv4 (`0x0800`), ARP (`0x0806`), LLDP (`0x88cc`), and BDDP (`0x8942`). All 80 464 default entries execute the explicit action `OFPAT_OUTPUT`: `OFPP_CONTROLLER` (controller port `0xfffffffd`).
2) `(in_port, eth_src, eth_dst)`: Reactive MAC-pair forwarding rules (24 437 entry observations). All 24 437 entries specify `OFPAT_OUTPUT` to physical switch data ports.

Not a single entry in any flow-statistics reply matches on source IP, destination IP, transport protocol, or transport port numbers. A 5-tuple flow—the fundamental unit labelled in InSDN's CSV—is never represented as an individual counter within the controller's flow table.

To verify whether transient 5-tuple entries could have been installed and deleted between polling intervals, we analyzed all 3 827 FlowMod messages transmitted by the ONOS controller. 3 811 FlowMods match strictly on `(in_port, eth_dst, eth_src)`, 16 match on `eth_type`, and exactly zero FlowMods specify IP addresses or transport ports. These FlowMods are consistent with an Ethernet learning switch; IPv4 reaches the controller through the punt rule described below.

C. Dynamics of the Controller PacketIn Stream

The controller's visibility of data traffic comes from PacketIns generated by ONOS's default punting rules and by reactive-forwarding misses. The flow-table decode shows an `eth_type=0x0800`→CONTROLLER rule in all 44 captures: all 20 116 observations have priority 5, below the MAC-pair rules (all 24 437 observations at priority 10), while the ARP, LLDP and BDDP punt rules have priority 40 000. Only 3 827 FlowMods occur in the 44 captures, and five captures contain none (Metasploitable DDoS SYN and TCP, and the three OVS web-attack captures); their flow-statistics replies nevertheless show MAC-pair entries (4–73 observations per capture), which were installed before the capture started. We did not establish why packets that such entries should cover still reach the controller; candidates are traffic that matches no installed pair (spoofed or changing address combinations, one-way traffic) and ONOS applications that request packets, but we tested none of them. In Table IV and Fig. 2, we pair target-side packet counts with controller-side PacketIn volume across all attack scenarios.

In 9 of 12 attack scenarios, the controller receives between 0.68 and 1.29 PacketIns per data packet recorded at the target. Two Metasploitable-2 scenarios show 3.9 and 15.7. The OVS DDoS scenario shows 0.08 (6.5M target packets against 541k PacketIns), consistent with an untested hypothesis of control-channel saturation under a spoofed flood; we did not test this directly. The two sides are separate recordings and their alignment is unverified, so ratios are indicative, not exact per-packet counts. The Normal group has no target-side captures, so we have no such ratio for benign traffic. Because each PacketIn carries the packet's headers, the 5-tuple is visible to the controller per packet, but never as a per-flow counter in the switch flow table.

Software flow reconstruction from PacketIns. Because PacketIns carry packet headers, a monitor could try to reassemble 5-tuple flow records in software. We implemented a custom streaming flow reassembler incorporating CICFlowMeter-inspired expiration rules (symmetric 5-tuple key, 120 s inactivity

TABLE I
FLOWS REASSEMBLED FROM PACKETINS (120 S TIMEOUT, FIN/RST TERMINATION) AGAINST THE BENCHMARK CSV FOR ONE CAPTURE PER SCENARIO. CORRELATIONS AND THE PACKET RATIO USE ONLY MATCHED 5-TUPLES AND ARE OMITTED (–) BELOW 30 MATCHES. RATIO = RECONSTRUCTED OVER HOST-SIDE PACKET TOTAL.

Scenario	Flows	1-pkt %	Matched	Matched %	ρ_{pkt}	Ratio	ρ_{dur}
OVS DDoS	17,566	100.0	0	0.0	–	–	–
OVS Probe	11,293	97.3	0	0.0	–	–	–
OVS Web	61	37.7	8	19.7	–	–	–
MS2 DoS	29,818	7.1	494	1.7	0.24	0.38	-0.27
MS2 Probe	13,589	0.6	13,214	97.7	0.86	1.00	-0.11
Normal (1 of 11)	501	35.3	140	65.9	0.94	0.45	0.98

timeout, TCP FIN/RST termination) and ran it on one capture per scenario, comparing the result with the benchmark CSV (Table I, `packetin_reconstruction.json`). Three observations stand out. (i) In spoofed volumetric attacks and scans almost every reconstructed flow is a single packet (OVS DDoS 100.0%, OVS Probe 97.3%). The target-side traces show the same structure (6 510 923 packets over 6 508 402 distinct 5-tuples for OVS DDoS, Table IV), so this reflects the traffic and not information lost in the control channel. The benchmark CSV, by contrast, contains no single-packet flow in any of the six subsets examined, and for the two OVS captures we found no 5-tuple in common with it (Protocol and ports are zero in all OVS DDoS rows; for OVS Probe the cause is not established), so no accuracy statement is possible there. (ii) Where flows can be matched, rank agreement of packet counts is high on probing ($\rho = 0.86$) and benign traffic ($\rho = 0.94$) but weak on the DoS capture ($\rho = 0.24$, covering 1.7% of its reconstructed flows), and reconstructed packet totals show 38% (DoS) and 45% (benign) indicative packet correspondence with host-side totals in the examined unsynchronized recordings, so the controller sees only a fraction of the packets of the flows it reassembles; on probing the totals agree (ratio 1.00). (iii) Durations are not reliably recovered: rank correlation is 0.98 on the benign capture, but -0.27 on DoS and -0.11 on probing, where almost all durations are near zero. We did not test why durations diverge under flooding (queueing or buffering of PacketIns is one candidate). The evidence shows only that, with this reassembler and these captures, software reconstruction recovers packet counts and durations in some scenarios and not in others. Matched flows are a subset of the reconstructed flows (Table I), the benign row is one of eleven captures, and the CSV is a separate recording whose alignment with the controller capture is unverified.

V. CONTROLLER-OBSERVABLE INTRUSION DETECTION

A. Streaming OpenFlow 1.3 Parser

We implemented a streaming OpenFlow 1.3 parser designed to reassemble TCP streams on port 6653. The parser handles message boundary segmentation and employs heuristic desynchronization recovery. If a message header is corrupted or desynchronized, the parser scans forward until it discovers two consecutive valid OpenFlow headers. Across all 44 captures, the parser processed 7 381 679 messages, requiring only 134 resynchronizations and discarding 12 217 bytes in total.

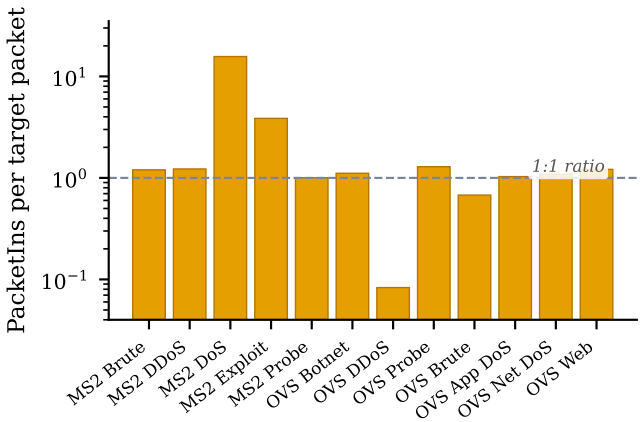


Fig. 2. PacketIns received by the ONOS controller per packet seen at the target host across InSDN attack scenarios (MS2: Metasploitable-2 target, OVS: OVS switch target). The dashed line marks parity (one PacketIn per target packet). Data: `gap_attack_v2.json`.

B. Feature Engineering

For consecutive non-overlapping windows of duration $W \in \{1\text{s}, 5\text{s}\}$, we extract 22 controller-observable features:

- **Control Message Counts (5):** Total counts of PacketIn, PacketOut, FlowMod, FlowRemoved, and FlowStats replies.
- **Header Diversity and Distributions (9):** Distinct source MACs, destination MACs, source IPs, destination ports; fractions of TCP, UDP, ICMP, and ARP traffic; and mean frame length.
- **Protocol and Entropy Metrics (3):** TCP SYN-only packet fraction, Shannon entropy of source IPs, and Shannon entropy of destination ports.
- **Flow Table Counter Deltas (5):** Number of active entries reported in statistics replies, sum and maximum of packet deltas, and sum and maximum of byte deltas between consecutive polls.

To capture temporal dynamics, the feature vector concatenates the current window with three preceding windows ($H = 4$), applying a $\log(1 + x)$ transformation to normalize dynamic ranges.

C. Strict Ground-Truth Window Labeling

Assigning ground-truth labels to controller windows requires precision. Marking every window within an attack capture as malicious would mislabel idle periods. Instead, we match the 5-tuple header of every PacketIn message against InSDN's ground-truth CSV flow records.

A window is labelled as an attack window if and only if it contains at least one PacketIn matching a known attack flow. In attack captures, windows containing unmatched IP traffic are marked as ambiguous and excluded from training and evaluation. Benign captures retain all windows. As shown in Table III, this strict protocol yields a sparse set of verified attack windows for low-volume classes (e.g., 6 windows for Brute Force, 12 for Exploit).

TABLE II
STRICTLY LABELLED CONTROLLER WINDOWS (W=1 S) PER DEPLOYMENT AND CLASS. A CLASS WITH ZERO ATTACK WINDOWS CONTRIBUTES ONLY ATTACK-FREE WINDOWS: OVS DDOS AND OVS BRUTE-FORCE HAVE NO CSV-MATCHED FLOWS (SECTION V). PROBE SUPPLIES 75.9% OF THE 2,589 POOLED ATTACK WINDOWS.

Deployment	Class	Captures	Windows	Kept	Attack win.
Metasp.	Brute force	2	45	33	6
Metasp.	DDoS	3	290	101	55
Metasp.	DoS	2	333	308	133
Metasp.	Exploit	4	88	71	12
Metasp.	Probe	2	1,989	1,956	1,922
OVS	Botnet	1	160	152	83
OVS	Brute force	2	559	326	0
OVS	DDoS	3	209	83	0
OVS	DoS	9	528	497	323
OVS	Probe	2	3,979	965	43
OVS	Web	3	636	275	12
Normal	Normal	11	33,283	33,283	0

Two consequences of this construction shape everything that follows. First, a label requires the CSV to carry a usable 5-tuple. In `OVS.csv` all 48 413 DDoS rows have Protocol, source port and destination port set to 0, so no PacketIn can match them, and the three OVS DDoS captures contribute no attack windows. The two OVS brute-force captures also contribute none: their CSV rows target the container address 172.17.0.2 and we found no PacketIn with a matching 5-tuple (the cause is not established). Second, the labelled attack windows are dominated by probing: Table II shows that probe windows are 75.9% of all 2 589 attack windows, and the two Metasploitable-2 probe captures alone supply 1 922. Pooled scores are therefore close to probe-detection scores, and per-class results must be read with these counts in mind. By construction, attack windows are windows in which at least one CSV-matched flow crossed the controller; this selection may favour easy windows, and we cannot quantify that effect here.

D. Evaluated Models and Baselines

We evaluate several detector architectures:

- **Multi-Layer Perceptron (MLP):** A fully connected network (64–32 units, ReLU activation, dropout rate 0.2, class-weighted cross-entropy loss, Adam optimizer, 40 epochs).
- **Linear Classifier:** Logistic regression with L2 regularization operating on the same historical feature inputs.
- **Single-Window MLP:** An MLP evaluating only the current 1 s window without temporal history.
- **Feature-Ablated MLPs:** Models trained exclusively on PacketIn-derived features or flow-statistics features.
- **Threshold Heuristics:** Univariate decision thresholds fitted on training folds (PacketIn rate, normalized burst rate, source-IP entropy, destination-port entropy).
- **Gradient Boosted Trees (XGBoost):** An optimized XGBoost baseline trained on the GPU using the same window features.
- **Host-Side Flow Baseline:** XGBoost trained on InSDN's 78 numeric CICFlowMeter features, which include the source and destination port numbers, under standard 70:30 flow splits. The audit of Section VIII repeats the split without ports, IP addresses, Flow ID and timestamps (77 features).

E. Validation Protocols

All evaluations enforce capture-level separation: no capture file appears in both training and testing sets. Metrics are reported across 10 random seeds as mean with 95% t-interval over seeds. For P1 each seed draws a new capture-to-fold assignment and a new network initialization, so this interval reflects both and not the sampling of captures; the capture-level cluster bootstrap in Table VI addresses the latter. The three detection protocols are:

- **P1 (Capture-grouped 5-fold CV):** Captures are partitioned into five folds, stratified by class.
- **P2 (Withheld attack families):** Entire attack families are held out (DoS and DDoS pooled as flooding); benign captures are split 70:30 between training and testing.
- **P3 (Cross-deployment):** Training on Metasploitable-2 sessions and testing on OVS sessions, and vice versa; benign captures are split 70:30 as in P2.

We report Balanced Accuracy (BAcc), Recall, and False Positive Rate (FPR). The evaluation-validity audit of Section VIII uses its own protocols A–E.

VI. EMPIRICAL DETECTION RESULTS

A. Performance on Unseen Captures

Table V presents detection results on unseen captures under 5-fold cross-validation (W=1 s). Across 10 seeds (Section V), the controller-observable MLP achieves a balanced accuracy of 0.946 ± 0.016 with a false positive rate of 0.001. Logistic regression performs competitively at 0.939, and the single-window MLP reaches 0.934, indicating that temporal history provides marginal separation gain in this feature space.

In contrast, standard SDN heuristics perform poorly: the PacketIn-rate threshold achieves only 0.616 balanced accuracy (FPR 0.561), failing because benign interactive sessions also generate bursty PacketIn streams. Entropy thresholds perform substantially better (0.839–0.852). Ablation confirms that PacketIn features drive performance (0.943 alone), while flow-statistics features alone achieve only 0.699 with an FPR of 0.284, consistent with coarse MAC-pair aggregation.

While Table V reports unweighted fold-averaged variation across seeds (± 0.016; each seed redraws the capture-to-fold assignment and the initialization), a cluster bootstrap resampling over captures (Table VI) measures capture-level variance, yielding a pooled BAcc of 0.984 [0.926, 0.995] for the MLP. The divergence between fold-averaged BAcc (0.946) and pooled window BAcc (0.984) is direct evidence of class and window concentration: because probe windows represent 74.2% of strictly labeled attack windows and achieve 0.997 recall, pooling windows across folds heavily weights probe detection. In contrast, unweighted fold averaging weights minority classes and smaller folds equally, pulling the mean score to 0.946. A label-shuffle sanity check produces a balanced accuracy of 0.512 ± 0.054 at 1 s and 0.508 ± 0.067 at 5 s, confirming chance level.

TABLE III
CONTROLLER CAPTURES (1 S WINDOWS). UNDER STRICT LABELS, AMBIGUOUS WINDOWS OF ATTACK CAPTURES ARE EXCLUDED ("KEPT").

Class	Captures	Windows	Kept	Attack win.	PacketIns
Normal	11	33,283	33,283	0	789,569
DoS	11	861	805	456	845,362
DDoS	6	499	184	55	1,340,347
Probe	4	5,968	2,921	1,965	621,986
Brute force	4	604	359	6	5,564
Exploit	4	88	71	12	332
Web	3	636	275	12	1,199
Botnet	1	160	152	83	1,002

TABLE IV
TARGET-SIDE TRAFFIC VS. CONTROLLER OBSERVATIONS PER SCENARIO (RATIO = PACKETINS PER TARGET PACKET).

Group	Scenario	Target pkts	5-tuples	PacketIns	Ratio
Metasp.	Brute force	1,311	295	1,574	1.20
Metasp.	DDoS	652,495	651,083	799,253	1.22
Metasp.	DoS	13,078	989	205,198	15.69
Metasp.	Exploit	86	18	332	3.86
Metasp.	Probe	507,650	51,150	510,194	1.01
OVS	Botnet	902	164	1,002	1.11
OVS	DDoS	6,510,923	6,508,402	541,094	0.08
OVS	Probe	86,705	79,021	111,792	1.29
OVS	Brute force	5,892	1,112	3,990	0.68
OVS	App. DoS	299,625	41,738	308,681	1.03
OVS	Net. DoS	304,886	20,979	331,483	1.09
OVS	Web	987	176	1,199	1.21

TABLE V
UNSEEN CAPTURES, GROUPED 5-FOLD CV, $W=1$ S, STRICT LABELS, 10 SEEDS (MEAN±95% CI OVER SEEDS).

Method	BAcc	Recall	FPR
MLP, all features	0.946±0.016	0.893	0.001
MLP, PacketIn features only	0.943±0.015	0.887	0.002
Logistic regression (history)	0.939±0.016	0.887	0.009
MLP, single window	0.934±0.015	0.870	0.003
Thr. source-IP entropy	0.852±0.029	0.763	0.059
Thr. dst-port entropy	0.839±0.018	0.736	0.059
MLP, flow-stats features only	0.699±0.036	0.683	0.284
Thr. PacketIn burst	0.643±0.008	0.912	0.627
Thr. PacketIn rate	0.616±0.011	0.793	0.561
MLP, within attack sessions	0.942±0.016	0.893	0.008
macro-averaged over captures	0.906±0.009	–	–
Trivial: "any IP PacketIn" (same windows)	0.995	–	–
macro over captures	1.000	–	–

B. Withheld Attack Families and Cross-Deployment Transfer

Under leave-one-attack-family-out testing (Table VII), the controller-side MLP reaches 0.936 on floods, 0.997 on probes and 0.904 on web attacks, above the best single-feature threshold in those families; these scores are subject to the session confound of Section VII. On sparse minority classes, thresholds or linear models match or exceed it (e.g., botnet threshold 0.965 vs. MLP 0.932; logistic regression 0.997 on exploit). The brute-force, exploit and web families rest on only 6, 12 and 12 attack windows (Table II), so their scores carry uncertainty that the seed interval does not capture. In cross-deployment evaluations (Table VIII), training on OVS sessions and testing on Metasploitable-2 yields 0.990 balanced accuracy, while the reverse direction achieves 0.905 (with logistic regression reaching 0.928). The OVS→Metasploitable-2 direction is scored on 2 128 attack windows, 1 922 (90%) of them probe windows; the reverse direction is scored on 461 OVS attack windows (botnet 83, DoS 323, probe 43, web 12). Extending the window length to $W=5$ s yields consistent performance (MLP 0.958, logistic regression 0.961, XGBoost 0.952).

C. Comparison with Host-Side Flow Baseline

For contrast, applying XGBoost to InSDN's host-side CICFlowMeter features (343 889 rows, 78 numeric features including the two port numbers) under standard randomized 70:30 flow splits yields a balanced accuracy of 0.9998 ± 0.00003. This near-perfect accuracy should not be read as evidence of detection ability: Section VIII shows that it does not survive address-disjoint evaluation on this dataset.

VII. VALIDITY ANALYSIS: DISENTANGLING ATTACKS FROM SESSION ARTIFACTS

A. The Trivial IP PacketIn Rule

In the distributed captures, benign and attack traffic were recorded in separate sessions, and the attack sessions contain no identifiable concurrent benign background traffic (see the window counts below). Strict labelling therefore confounds session type with the label.

When we evaluate the MLP strictly within attack sessions, it achieves a pooled balanced accuracy of 0.942 (macro-averaged 0.906). However, a trivial baseline rule—"classify window as attack if it contains at least one IP PacketIn"—achieves a pooled balanced accuracy of 0.995 and a macro-averaged score of 1.000 across the exact same windows (Fig. 3). This 0.995 score reflects the labeling protocol combined with session separation: because the ambiguity mask excludes attack-session windows containing unlabelled IP traffic, the surviving attack-free windows contain almost no IP traffic. Of the 2 178 attack-free windows kept in attack sessions, 1 858 are empty (no PacketIn at all) and are excluded from this metric; of the remaining 320, 317 contain no IP PacketIn (ARP, LLDP or other non-IP frames). The 0.995 therefore rests on 2 589 attack windows and 320 attack-free windows. Thus, any rule checking for the presence of IP packets achieves near-perfect separation inside attack sessions without learning any attack characteristics.

Operational Evaluation across Benign Traffic. While the trivial rule exploits this non-interleaved benchmark artifact within attack sessions, evaluating it against operational benign traffic (the 11 Normal sessions, 33,283 windows) exposes its failure mode (`operational_stream_evaluation.json`): because 77.5% of normal-session windows contain IP PacketIns (25,810 of 33,283 windows), the trivial rule flags 77.5% of normal-session traffic as attacks (normal-session FPR = 0.775). Evaluated on the full strict labeled window set (38,050 windows: 2,589 attack, 35,461 pooled benign), every attack window contains an IP PacketIn by construction of strict labels (Recall = 1.000), but its overall benign FPR of 72.8% drops balanced accuracy to 0.6360 (and to 0.4889

TABLE VI
POOLED WINDOW-COUNT BACC WITH A CAPTURE-LEVEL CLUSTER BOOTSTRAP (2000 RESAMPLES, STRATIFIED BY CLASS).

Method	Pooled BAcc	95% CI
MLP, all features	0.984	[0.926, 0.995]
Logistic regression	0.979	[0.921, 0.991]
MLP, single window	0.979	[0.905, 0.993]
Thr. source-IP entropy	0.923	[0.783, 0.959]
Thr. dst-port entropy	0.926	[0.792, 0.953]
Thr. PacketIn rate	0.467	[0.245, 0.817]

TABLE VII
WITHHELD ATTACK FAMILY (BACC, W=1 S, STRICT LABELS). "BEST THR.": BEST OF FOUR THRESHOLDS, CHOSEN PER FAMILY.

Withheld family	MLP	Logistic	Best thr.
Flood (DoS+DDoS)	0.936±0.011	0.953	0.814
Probe	0.997±0.000	0.988	0.970
Web	0.904±0.020	0.909	0.699
Brute force	0.891±0.029	0.913	0.888
Exploit	0.929±0.037	0.997	0.929
Botnet	0.932±0.024	0.803	0.965

across the raw unmasked 42,099-window stream). In contrast, the controller-observable MLP avoids triggering on benign IP traffic, achieving a false positive rate of only 0.001 on normal sessions and 0.946 balanced accuracy overall (Table V). However, as our session probes demonstrate below, avoiding simple heuristic triggers does not mean the MLP has learned invariant attack semantics: classifiers trained on residual traffic after removing attack-matched PacketIn messages still predict session origin with 0.940 balanced accuracy (residual windows may still contain unmatched attack traffic, Table IX). Thus, non-interleaved benchmark recordings make it difficult to untangle genuine attack detection from controller-level session fingerprints.

B. Host-Identity Baseline

To bound how much of the controller-side score can be explained by *which hosts appear*, without any traffic statistics, we evaluate two rules learned on the training captures of each fold (same folds, labels, windows and seeds as Table V). Rule R1 predicts attack if the window contains a PacketIn source address that appeared in training attack windows and in no training attack-free window; rule R2 predicts attack if it contains any source address seen in training attack windows (`session_identity_baseline.json`). Table XIII shows that R1 reaches a balanced accuracy of 0.734 over all strictly labelled windows, with recall limited to 0.472 because the source addresses of spoofed floods never repeat across captures, and that R2 (0.680) is dominated by false positives on benign sessions. Within attack sessions R2 is again near-perfect (0.994), like the any-IP rule.

Host identity therefore explains a substantial part of the separation but not all of it: the MLP exceeds R1 by 0.21 balanced accuracy, so its score is not reducible to address memory. However, the session probe below still separates sessions at 0.940 after attack-matched PacketIns are removed, so we cannot attribute the remaining margin to attack semantics rather than to other session fingerprints such as traffic mix or timing. The rules are simple; a stronger identity model could score higher, so these numbers are lower bounds on what identity can explain.

TABLE VIII
CROSS-DEPLOYMENT TRANSFER (BACC, STRICT LABELS).

Train → test	MLP	Logistic	Best thr.
Metasploitable-2 → OVS	0.905±0.021	0.928	0.866
OVS → Metasploitable-2	0.990±0.001	0.985	0.947

TABLE IX
SESSION PROBE: PREDICTING BENIGN VS. ATTACK *session* FROM PACKETIN FEATURES AFTER REMOVING ATTACK-MATCHED PACKETINS (≥5 REMAINING PER WINDOW). THE RESIDUAL CAN STILL CONTAIN UNLISTED ATTACK TRAFFIC (D1: 424 OF 518 ATTACK-SESSION WINDOWS CONTAIN UNMATCHED-IP PACKETINS), AND D2 RESTS ON 6 WINDOWS FROM 2 CAPTURES.

Variant	BAcc	Majority
All windows (D1)	0.940±0.005	0.500
Attack-free windows only (D2)	0.950±0.031	0.500

C. Session Probes

To test whether classifiers recognize session identity rather than attack signatures, we trained an MLP on windows where all attack-matched PacketIns were excised (Table IX). The model predicts whether a window originates from an attack session with 0.940 balanced accuracy (D1). When restricted to attack-free windows (D2), it achieves 0.950. However, 82% of these residual windows contain unmatched IP traffic that likely represents unlabelled attack remnants, and D2 rests on only 6 windows from 2 captures. These findings demonstrate that in non-interleaved benchmark recordings, attack detection cannot be cleanly disentangled from session fingerprinting.

VIII. EVALUATION-VALIDITY AUDIT: NETWORK PARTITIONING PROTOCOLS AND ADDRESS GENERALIZATION

A. Experimental Design Across Two Benchmarks

To evaluate leakage in flow-based evaluations, we implemented a standardized evaluation pipeline applied to two prominent benchmarks: InSDN (343 889 flows) and an external reference dataset, CICIDS2017 (2 100 814 flows). We evaluate an identical GPU-accelerated XGBoost classifier across distinct protocols using 10 random seeds (Table X and Fig. 4):

- **Protocol A (Random 70:30 Flow Split):** Standard randomized flow-level partition. Excludes all IP addresses, ports, Flow IDs, and timestamps (retaining 77 numeric features).
- **Protocol B (Source-IP-Disjoint Split):** 70:30 partition grouped by source IP. No source address appears in both training and test sets.
- **Protocol C (Time-Ordered Split):** Deterministic 70:30 chronological partition per capture file.

TABLE X
EVALUATION-VALIDITY AUDIT: BALANCED ACCURACY (BENIGN VS. ATTACK) OF THE SAME XGBOOST (GPU) ON THE 77 NUMERIC CICFLOWMETER FEATURES, SAME PIPELINE ON BOTH DATASETS, 10 SEEDS (MEAN±95% CI OVER SEEDS). A, B AND C USE NO IPS, PORTS OR TIMESTAMPS; D USES ONLY THOSE. ROWS: INSDN 343 889, CICIDS2017 2 100 814. C IS A SINGLE DETERMINISTIC SPLIT.

Protocol	InSDN	CICIDS2017
A. Random 70:30 flow split	0.9998	0.9917±0.0001
B. Source-IP-disjoint split (70:30 by source IP)	0.907±0.085	0.641±0.056
C. Per-file time-ordered 70:30 (deterministic)	0.9997	0.7187
D. IPs, ports and Flow ID only, random split	1.0000	0.9999
E. Split-A test rows duplicating a train row (%)	48.7	20.3

TABLE XI
FEATURE-RECIPE REPLICATION ON INSDN (ONE XGBOOST CLASSIFIER, GPU, 80:20, 10 SEEDS, MEAN±95% CI): BINARY BALANCED ACCURACY AND 8-CLASS ACCURACY UNDER A RANDOM FLOW SPLIT AND A SOURCE-IP-GROUPED SPLIT. RECIPES APPROXIMATE FEATURE CHOICES DESCRIBED IN PUBLISHED INSDN PAPERS; THEY ARE NOT THOSE PAPERS' MODELS OR CODE. GROUPING BY SOURCE IP ALSO CHANGES THE ATTACK-CLASS MIX BETWEEN TRAIN AND TEST, SO GROUPED VALUES ARE A STRESS TEST.

Recipe	BAcc random	BAcc grouped	8-cls random	8-cls grouped
Numeric features only	0.9998±0.0000	0.943±0.057	0.9991±0.0001	0.788±0.186
Numeric + ports	0.9998±0.0000	0.856±0.151	0.9998±0.0000	0.687±0.241
Numeric + identifier columns	1.0000±0.0000	0.941±0.088	0.9999±0.0000	0.804±0.187
Top-15 by MI, selected before split	1.0000±0.0000	0.941±0.088	0.9999±0.0000	0.835±0.175

- **Protocol D (Identifier-Only Split):** Random 70:30 split trained exclusively on IP addresses, port numbers, and Flow IDs.
- **Protocol E (Redundancy Audit):** Measurement of the percentage of test-set rows that possess an exact feature duplicate in the training set under Protocol A.

B. The Generalization Collapse on Unseen Addresses

Under Protocol A (random flow split), XGBoost achieves near-perfect balanced accuracy: 0.9998 on InSDN and 0.9917± 0.0001 on CICIDS2017. However, under Protocol B (source-IP-disjoint split), performance on InSDN drops by nine percentage points to 0.907 ± 0.085, with individual splits swinging from 1.00 down to 0.66. Across the 10 random seeds on InSDN, the balanced accuracy distribution is widely dispersed: five seeds score near 1.00 (0.996, 0.998, 0.999, 1.000, 1.000), two score between 0.85 and 0.94 (0.859, 0.939), and three drop below 0.84 (0.660, 0.786, 0.837), consistent with the acute host concentration documented below. On CICIDS2017, the drop is catastrophic: performance collapses by 35 percentage points to 0.641 ± 0.056. A per-file time-ordered split shows no drop on InSDN (0.9997) but falls to 0.719 on CICIDS2017 (deterministic chronological split).

Under Protocol D, a classifier supplied exclusively with network identifiers (IP addresses, ports, and Flow ID) achieves 1.0000 on InSDN and 0.9999 on CICIDS2017. Network identifiers alone therefore separate the classes in these benchmarks; this does not show that models trained on flow features use them, only that they could.

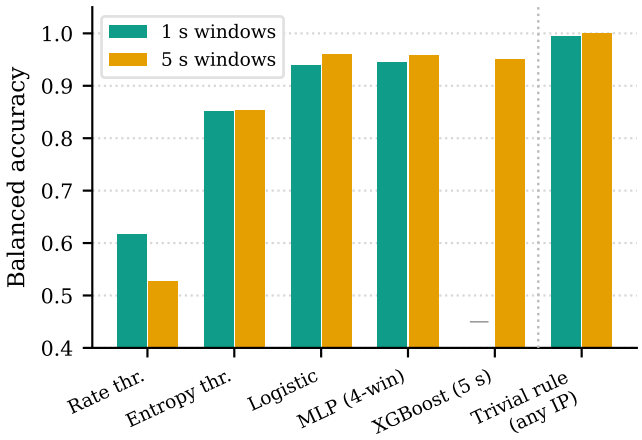


Fig. 3. Balanced accuracy on unseen captures under capture-grouped CV across 10 seeds (XGBoost evaluated at 5 s). Separated by a dotted line on the right, the trivial rule ("at least one IP PacketIn") matches or outperforms all machine learning models within attack sessions.

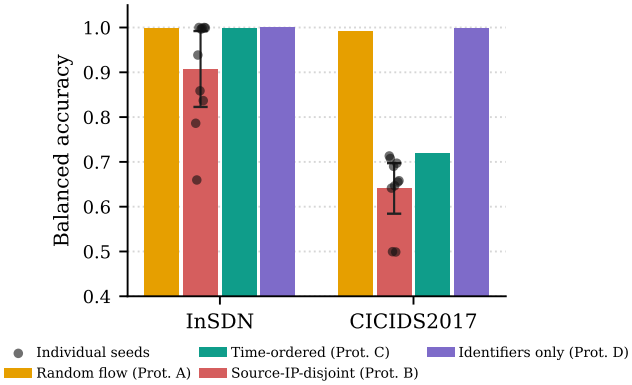


Fig. 4. Balanced accuracy across four evaluation protocols on InSDN and CICIDS2017 (10 seeds; dots represent individual source-IP group splits). When source-IP groups are held out, performance degrades sharply, and identifier-only models achieve near-perfect scores.

C. Construct Validity: Source IP Is Not Host Identity

A foundational construct-validity flaw in published NIDS literature is the conflation of source IP addresses with physical host identity. InSDN contains 122,868 distinct source IP addresses; however, 122,307 of these source IPs appear in exactly one flow record. A forensic breakdown reveals that 121,938 of these single-flow addresses (99.7%) belong to the `DDoS` class. We verified this empirical structure directly in the raw packet captures (`h4-DDoS-ICMP.pcap` and `metasploitable-2_ddos-icmp.pcap`): 100.0% of the underlying frames originate from exactly two physical source MAC addresses (`00:00:00:00:00:01` and `00:00:00:00:00:04` in the OVS testbed; `00:0c:29:92:12:c4` and `00:0c:29:fa:dd:2a` in Metasploitable), but carry randomized IPv4 source addresses across arbitrary subnets generated by an automated packet generator (`hping3 --rand-source`). In the benchmark CSV, 121 920 of these 121 938 records (99.985%) contain zero forward packets and a median duration

TABLE XII
TARGETED LITERATURE AUDIT OF REPRESENTATIVE SDN INTRUSION DETECTION EVALUATIONS ON BENCHMARK DATASETS (12 PAPERS EXAMINED, 10 READ IN FULL TEXT; DETAILS AND QUOTATIONS IN `LITERATURE_CHECK.JSON`). NONE OF THE INSDN PAPERS READ IN FULL REPORTS A HOST-GROUPED OR CAPTURE-GROUPED SPLIT, AND THREE USE IDENTIFIER ATTRIBUTES AS INPUT FEATURES.

Paper	Venue	Dataset	Evaluated Model	Split Protocol	Identifiers Used?	Methodological Findings / Red Flags
Towhid & Shahriar [35]	NOMS 2023	InSDN	RF / LSTM	Random 70:30	Excluded	Divides all flows 70:30; no host/session grouping; offline Scapy features
Ataa et al. [4]	Sci. Rep. 2024	InSDN	CNN-LSTM / Transf.	Shuffled (ratio unstated)	Excluded	Shuffled flows; no train/test ratio or grouping stated; single run
Abdallah et al. [36]	ACM ARES 2021	InSDN	Hybrid CNN-LSTM	Unverified (paywall)	Unverified	Publisher access restricted; split details unverified
Elsayed et al. [37]	ASEJ 2023	InSDN / ToN-IoT	SATIDS (LSTM)	Unverified (paywall)	Unverified	Publisher access restricted; split details unverified
Saykat et al. [6]	Sci. Rep. 2026	InSDN / CIC-17	OptiXGB + WGAN-GP	80:20 sample-level	Src/Dst Port kept	Preprocessing leakage controlled, but the split has no host grouping; Dst Port is the major split of the surrogate tree
Mujahid et al. [10]	PeerJ CS 2025	InSDN / Kaggle	RF + Deep Extractor	train_test_split (ratio unstated)	Unstated	scikit-learn row-level split; no grouping; row count mismatch unexplained
Mehmood et al. [7]	PLOS ONE 2025	InSDN / CICDDoS	CNN-MLP	Unstated	Dst Port kept	No split protocol stated; Dst Port is the strongest negative SHAP contributor
Alqhatani et al. [38]	PLOS ONE 2025	UNSW / BoT-IoT	GRU-NTM	Chronological 80:20	Unstated	Chronological split, but the 20% is a validation set used for early stopping/tuning
Ashikuzzaman et al. [5]	arXiv 2026	InSDN	Stacking + LightGBM	80:20 (method unstated)	Flow ID, IPs, Ports	Top-15 mutual-information features include Flow ID, Src/Dst IP, Src/Dst Port; SHAP ranks Flow ID first
Malik et al. [39]	Unreviewed ms. 2026	InSDN	BiLSTM + RFE	Stratified 5-fold CV	Unstated	Stratified 5-fold drops accuracy to 81.60%; attributed to split variance
Rohith et al. [40]	Sci. Rep. 2026	InSDN	Quantum + MSDC Net	80:20 (method unstated)	Unstated	Four split ratios tried; 80:20 chosen from the reported test results; no grouping
Saleh & Hamad [41]	JESA 2026	SICA / InSDN	DNN / LSTM	70:20:10 (method unstated)	Excluded	Removed Flow ID/IPs, yet InSDN scores 99.61% (split presumably random)

TABLE XIII
HOST-IDENTITY RULES AGAINST THE CONTROLLER-SIDE MLP, STRICT LABELS, $W{=}1$ S, CAPTURE-GROUPED 5-FOLD CV, 10 SEEDS (MEAN±95% CI OVER SEEDS). RULES ARE LEARNED ON TRAINING CAPTURES ONLY AND USE NO TRAFFIC STATISTICS.

Method	BAcc	Recall	FPR
All strictly labelled windows			
MLP, all features (Table V)	0.946±0.016	0.893	0.001
R1: attacker-only IP memory	0.734±0.033	0.472	0.004
R2: any attack IP	0.680±0.009	0.997	0.637
Within attack sessions (non-empty windows)			
MLP, all features	0.942±0.016	0.893	0.008
R1: attacker-only IP memory	0.736±0.032	0.472	0.000
R2: any attack IP	0.994±0.003	0.997	0.008

TABLE XIV
EVALUATION-VALIDITY AUDIT ACROSS NETWORK PARTITIONING PROTOCOLS ON INSDN (77 NUMERIC CICFLOWMETER FEATURES, IDENTIFIERS EXCLUDED, GPU XGBOOST, 10 SEEDS, MEAN±95% CI).

Partitioning Protocol	Address Isolation	Balanced Accuracy	Min BAcc	Macro-F1
Protocol A: Random flow split	Overlapping	0.9998±0.0000	0.9997	0.9999±0.0000
Protocol B: Source-IP-disjoint	Disjoint Src IP	0.9074±0.0847	0.6596	0.8321±0.1883
Protocol B_{dst}: Destination-IP-disjoint	Disjoint Dst IP	0.8727±0.0881	0.5904	0.8141±0.1046
Protocol B_{pair}: (Src, Dst) Pair-disjoint	Disjoint (Src, Dst)	0.9891±0.0084	0.9630	0.9767±0.0193
Protocol $B_{nospoof}$: Non-spoofed Source IP	Disjoint (No Spoof)	0.8417±0.1114	0.6021	0.6503±0.2005

of 14 μs.

Consequently, standard source-IP-disjoint evaluation on raw InSDN does not test whether a model generalizes to novel physical endpoints; it merely partitions ephemeral spoofed headers from the same two physical machines across folds. The classifier trains on $\approx 85\,000$ spoofed DDoS flows and tests on $\approx 36\,000$ identical flows, artificially buoying the Protocol B score to 0.907.

To isolate this construct flaw, we executed a multi-protocol splitting audit (Table XIV, `multiprotocol_split_audit.json`):

- **Non-Spoofed Source-IP Disjoint (Protocol $B_{nospoof}$):** When single-flow spoofed DDoS traffic is removed and genuine non-spoofed traffic is evaluated on source-IP-disjoint splits, balanced accuracy collapses to 0.8417 ± 0.1114 (worst split 0.6021), and Macro-F1 plunges to 0.6503 ± 0.2005.
- **Destination-IP Disjoint Split (Protocol B_{dst}):** Grouping by destination IP yields 0.8727 ± 0.0881 (worst split 0.5904), proving that server-side address leakage also heavily inflates random splits.
- **Pair-Disjoint Split (Protocol B_{pair}):** Grouping by directed (Src, Dst) communication pairs yields 0.9891 ± 0.0084. Because attackers communicate with multiple victims and servers receive traffic from multiple sources, pair grouping permits host-level behavioral signatures to leak across partitions.

D. The Role of Data Redundancy and Class Coverage

Under Protocol E, we discover that 48.7% of random-split test rows in InSDN (and 20.3% in CICIDS2017) are exact duplicates of training rows. To test whether duplicate flows drive the random-split score, we deduplicated InSDN completely, reducing the dataset from 343 889 to 184,607 rows. On this deduplicated dataset, the random-split score remains unchanged at 0.9998. Thus, feature redundancy alone does not explain the performance gap.

To ensure that address grouping does not artificially penalize performance by withholding entire attack classes from training, we enforced a class-coverage constraint: holding out source-IP groups while guaranteeing that every attack class appears in the training partition. Under this controlled setting, balanced accuracy is 0.939 (0.812 after deduplication). Grouping by the 3-tuple `(src_ip, dst_ip, dst_port)` yields 0.961 (0.984 after deduplication). The triple-grouped split fragments traffic while retaining communication from the same attacker addresses on different ports across partitions, allowing the model to leverage address-level similarities without exact flow duplication.

E. Host Concentration and the Combinatorial Split Derivation

Why do source-IP-disjoint scores vary so widely? Only eight hosts account for 500 or more flows, yet these eight hosts encompass 215,404 of InSDN's 343,889 total rows. Among these eight dominant hosts, exactly three source attack-labelled flows: primary attacker `200.175.2.130` (110 799 flows, 32.2% of dataset), primary benign host `192.168.20.133` (53 790 flows, 15.6%), and secondary attacker `192.168.3.130` (37 374 flows).

By logging host assignments across all 10 evaluated Protocol B seeds (`host_assignment_audit.json`), we observe that the balanced accuracy dispersion correlates with dominant host allocations:

- Assigning primary attacker `200.175.2.130` to test (Seeds 0 and 1) elevates test row share to 43.7% and 57.1%, dropping balanced accuracy to 0.786 and 0.660.

TABLE XV
COMBINATORIAL DOMINANT HOST ALLOCATION ACROSS 50 PROTOCOL B SEEDS ON INSDN (GROUPSHUFFLESPLIT 70:30, $M = 122\,868, k = \lceil 0.3M \rceil = 36\,861$). CORRELATION BETWEEN TEST PARTITION ROW SHARE AND BALANCED ACCURACY: $r = -0.7713$ ($p = 5.61 \times 10^{-11}$).

Allocation Regime	Seeds (N)	Hypergeom. Theory	Empirical Pct	Mean BAcc	BAcc Range
Neither primary host withheld	24	49.0%	48.0%	0.9809	[0.9283, 1.0000]
Primary attacker withheld only	13	21.0%	26.0%	0.7513	[0.6596, 0.8172]
Primary benign withheld only	9	21.0%	18.0%	0.7566	[0.6300, 0.8591]
Both primary hosts withheld	4	9.0%	8.0%	0.7385	[0.5794, 0.8989]
Overall Population (50 Seeds)	**50**	**100.0%**	**100.0%**	**0.8614**$\pm$0.0377	**[0.5794, 1.0000]**

- Assigning primary benign host `192.168.20.133` to test (Seeds 4 and 7) drops balanced accuracy to 0.859 and 0.837 due to false positives on unseen benign activity.
- When neither primary host is withheld, performance is generally high: Seeds 3, 5, 6, 8, and 9 score between 0.996 and 1.000 (test row share 11.2%–22.1%). The one intermediate case is Seed 2 (BAcc 0.939, test share 22.2%), where neither primary host is in test. Host `192.168.20.1`, which yields an 18.8% FPR when held out alone (leave-one-source-IP-group-out, below), is in the test set of Seed 2 (and of Seed 0), consistent with the intermediate score.

Exact Combinatorial Derivation. In a 70:30 group partition of $M = 122\,868$ unique source addresses, scikit-learn's `GroupShuffleSplit` sets the test group count using the ceiling rule $k = \lceil 0.3 \times M \rceil = 36\,861$. The assignment is governed by the Hypergeometric distribution. Let E_{train} denote the event that both dominant hosts are retained in training. Its exact probability is:

$$P(E_{\text{train}}) = \frac{\binom{M-2}{k}}{\binom{M}{k}} = \frac{M-k}{M} \cdot \frac{M-k-1}{M-1} = 0.489991, \quad (3)$$

which is 49.0%. The symmetric probability that the primary attacker is assigned to test while the primary benign host is retained in train (E_{att}) is:

$$P(E_{\text{att}}) = \frac{k}{M} \cdot \frac{M-k}{M-1} = 0.210004, \quad (4)$$

which is 21.0%. The probability of withholding the primary benign host alone is likewise 21.0%, and both primary hosts are withheld with probability $\frac{k}{M} \cdot \frac{k-1}{M-1} = 9.0\%$.

To verify this formulation beyond small-sample 10-seed artifacts, we executed a 50-seed Monte Carlo evaluation (Table XV, `combinatorial_audit_50seeds.json`). Across 50 seeds, the empirical regime frequencies closely match hypergeometric theory: 48.0% neither withheld (theory 49.0%), 26.0% attacker only withheld (theory 21.0%), 18.0% benign only withheld (theory 21.0%), and 8.0% both withheld (theory 9.0%). The overall population balanced accuracy across 50 seeds is 0.8614 ± 0.0377 (ranging from 0.5794 to 1.0000). Crucially, the correlation between test partition row fraction and balanced accuracy is $r = -0.7713$ ($p = 5.61 \times 10^{-11}$). This strong association demonstrates that cross-seed variance in Protocol B reflects the stochastic allocation of dominant address groups under heavy-tailed group size distributions rather than unexplainable classifier noise.

To quantify generalization to individual endpoints, we conducted a Leave-One-Source-IP-Group-Out (LOSGO) evaluation, holding out one dominant address group at a time while training on all remaining records. When holding out the primary attacker (host 200.175.2.130), the model catches only 51.6% of its attack flows (DoS detection recall collapses to 34.7%). Conversely, when holding out the largest benign host (host 192.168.20.133), the false positive rate spikes to 37.9%. On CICIDS2017, which contains 48 source IPs (15 with ≥ 500 flows), holding out the primary attacker (172.16.0.1, 442 098 flows) results in an attack recall of only 28.3% and a benign false positive rate of 54.0%. Individual bot hosts held out in isolation achieve detection rates between 23.4% and 43.5%.

F. Sample Weighting: Gradient Analysis and Trade-Offs

To mitigate this collapse algorithmically, we investigated sample weighting across the 10 Protocol B seeds (Table XVI, `host_weighting_audit.json`). Consider a host group h with N_h training flows. In standard empirical risk minimization, the cumulative sample weight contribution of host h is:

$$\Omega_h = \sum_{i=1}^{N_h} w_i = N_h \cdot 1.0 = N_h. \quad (5)$$

The primary attacker (`200.175.2.130`, 110 799 flows) contributes $110\,799\times$ more sample weight mass than a single-flow address, inducing classifiers to overfit to its specific feature signature.

Under pure inverse-frequency weighting ($w_i \propto 1/N_h$), the host's cumulative sample weight contribution is $\Omega_h = N_h \cdot (1/N_h) = 1.0$, enforcing uniform host weighting. However, our empirical comparison in Table XVI demonstrates that pure inverse-frequency weighting catastrophically degrades performance on InSDN: mean balanced accuracy drops to 0.835 ± 0.158, and worst-case performance collapses to 0.3370 (in Seeds 4 and 7, where the primary benign host is in test). Because InSDN contains 121 938 single-packet spoofed DDoS addresses each assigned weight 1.0, pure inverse-frequency weighting inflates the cumulative weight of spoofed DDoS packets to $\approx 85\,000$, while downscaling the remaining benign hosts by $50\,000\times$, resulting in an operational false positive rate of 79.3% on benign traffic.

In Table XVI, clipped inverse-frequency (99% cap) yields scores bitwise identical to pure inverse-frequency. This is a direct mathematical consequence of InSDN's group distribution: 44.2% of all training records belong to single-flow source addresses ($N_h = 1$), assigning them sample weight $w_i = 1/1 = 1.0$. Because nearly half the dataset shares the theoretical maximum weight (1.0), the 90th, 95th, 99th, and 99.9th percentiles are all exactly 1.0. Upper-percentile clipping at 1.0 leaves the sample weights completely unmodified.

We note that while sample weights w_i scale the maximum gradient magnitude $g_i = w_i(p_i - y_i)$, realized gradient updates also depend on prediction error $(p_i - y_i)$. Once dominant attack flows are learned with high confidence, their residuals decay toward zero, moderating later updates. However, during early

TABLE XVI
COMPARISON OF SAMPLE-WEIGHTING STRATEGIES UNDER SOURCE-IP-DISJOINT EVALUATION (INSDN PROTOCOL B, 10 SEEDS, MEAN±95% CI). RATIO = TOTAL SAMPLE WEIGHT OF PRIMARY ATTACKER HOST RELATIVE TO A SINGLE-FLOW HOST. PAIRED DIFFERENCE FOR SQUARE-ROOT ATTENUATION VS. ERM: $\bar{\Delta} = +0.0238 \pm 0.0415$ ($p = 0.2267$).

Weighting Strategy	Sample Weight Ratio	BAcc	Worst Seed	Macro-F1
Unweighted ERM (Standard)	110,799 : 1	0.907±0.085	0.660	0.832±0.188
Square-root attenuation ($1/\sqrt{N_h}$)	333 : 1	0.931±0.064	0.730	0.835±0.149
Pure inverse-frequency ($1/N_h$)	1 : 1	0.835±0.158	0.337	0.706±0.193
Clipped inverse-frequency (99% cap)	1 : 1 (reg.)	0.835±0.158	0.337	0.706±0.193
Class-conditional ($1/N_{h,c}$)	Class-relative	0.826±0.158	0.342	0.685±0.202

boosting rounds, dominant address groups exert cumulative weight mass directly proportional to $N_h \times w_i$.

In contrast, square-root frequency attenuation ($w_i \propto 1/\sqrt{N_h}$) scales cumulative host weight contribution by $\Omega_h = N_h/\sqrt{N_h} = \sqrt{N_h}$. The primary attacker's weight dominance is attenuated from $110\,799\times$ down to $\sqrt{110799} \approx 332.8\times$. This provides a robust, variance-moderating trade-off: it prevents extreme gradient blowups from spoofed single-packet addresses while curbing dominant host overfitting. Across the 10 evaluated Protocol B splits, square-root attenuation raises mean balanced accuracy from 0.907 ± 0.085 to 0.931 ± 0.064, substantially mitigating worst-case performance: the worst-case split (Seed 1) jumps from 0.660 to 0.730 (+7.0 percentage points), Seed 0 (primary attacker withheld) jumps from 0.786 to 0.952 (+16.6 percentage points), and Seed 7 rises from 0.837 to 0.874. A paired per-seed test ($\Delta_i = \text{BAcc}_{\text{sqrt},i} - \text{BAcc}_{\text{unw},i}$) yields a positive mean paired difference $\bar{\Delta} = +0.0238 \pm 0.0415$ ($t = 1.298, p = 0.2267$), indicating that square-root attenuation acts as a variance-moderating tail hedge rather than an unconditional gain across all seeds.

G. Replication of Published Pipelines

We partially replicated the published SDNGuardStack pipeline [5] using our own implementation across 5 seeds. Under a random split, the model reproduces its published performance, achieving 0.9999 accuracy (published: 99.98%). However, under source-IP-disjoint evaluation, mean accuracy drops to 0.72 when selecting features strictly on training folds (per-seed: 0.24, 0.73, 1.00, 0.66, 0.99) and 0.78 when selecting on all rows (per-seed: 0.33, 0.99, 1.00, 0.62, 0.98); 95% confidence intervals span up to ±0.39. The metric is 8-class accuracy. This dispersion is consistent with InSDN's host concentration: a 20% host partition draws hosts with very unequal traffic volumes, the test row share varies from 8% to 40% across seeds, and the lowest score (0.24) occurs in the run with the largest test partition (40% of all rows), consistent with the test fold being dominated by a few large hosts while no test class is absent from training (`unseen_test_share` is 0.0 in all seeds). We did not test this attribution further.

As a second published pipeline we evaluated a *proxy* for Ataa et al. [4]. Their exact 48-feature set and architectures (a Transformer encoder and a CNN-LSTM) could not be reproduced from the text, so the proxy is a binary MLP on all numeric flow features; it is not their model, and their reported 99.02% is an 8-class accuracy that is not comparable with ours. Over 5 seeds the proxy reaches 0.9994 ± 0.0001 accuracy under a random split and 0.820 ± 0.202 under source-IP-disjoint evaluation (per-seed: 0.68, 0.71, 1.00, 0.71, 0.99; macro-F1 0.706 ± 0.299). The interval is again wide, consistent with the host concentration above. We make no claim about the original model.

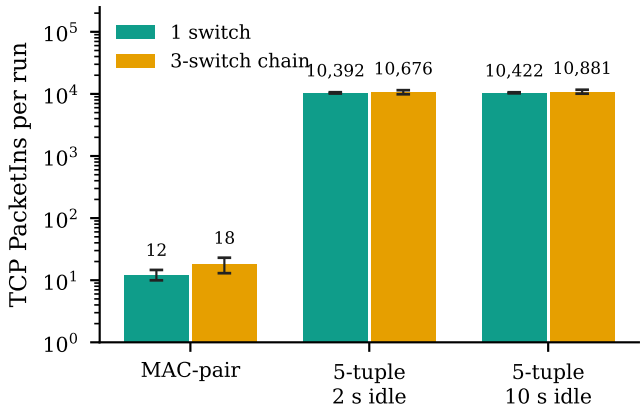


Fig. 5. TCP PacketIns per run (mean ± 95% CI, 10 runs per mode, log scale) under three forwarding modes, on one switch and on a three-switch chain. Data: `live_modes_v4_one10.json`, `live_modes_v4_linear3.json`.

IX. A FORWARDING-APPLICATION CASE STUDY

InSDN shows one forwarding configuration. To see whether what the controller observes depends on the forwarding application, we built a small testbed (Mininet, one Open vSwitch switch, six hosts; a Ryu OpenFlow 1.3 learning switch with a configurable entry granularity) and recorded its control channel with the same parser. The three modes install, for each learned destination, an entry matching (i) the MAC pair; (ii) the 5-tuple for TCP/UDP with a 10 s idle timeout; or (iii) the 5-tuple with a 2 s idle timeout applied to all entries. In the two 5-tuple modes, other IP traffic receives an (src, dst, protocol) entry and ARP an ARP-only entry, so a coarse entry for one protocol cannot match another. The controller polls flow statistics every second. Traffic is benign only and generated by the same program with the same random seeds in every mode (request timing still differs): five hosts repeatedly fetch files of 16 KB to 1 MB from an HTTP server (80%) or ping it (20%) with randomized uniform pacing (0.05–0.8 s) for 90 s (captures span ∼99 s including start-up). Each mode was run 10 times (seeds 0–9), reporting mean and 95% t-interval over runs (Table XVII); the 30 runs completed 26 891 requests, of which 41 failed (below). We repeated the study on a loop-free three-switch chain (Table XVIII, below).

Granularity sets the PacketIn volume. MAC-pair mode produced 12.3±2.3 TCP PacketIns per run, against 10 422±202 (10 s idle) and 10 392±249 (2 s idle) under per-flow forwarding: $665\times$ to $1\,752\times$ (10 s idle) and $668\times$ to $1\,780\times$ (2 s idle) more in each of the ten paired runs (larger in 10 of 10; two-sided sign-test $p = 0.002$). That is about 11–12 TCP PacketIns per completed request, so each connection is punted repeatedly; the most likely cause is packets that arrive before the FlowMod takes effect (we install entries without a barrier and send full

TABLE XVII
FORWARDING APPLICATION VS. WHAT THE CONTROLLER OBSERVES: BENIGN TRAFFIC FROM THE SAME GENERATOR AND SEEDS (5 HOSTS → 1 HTTP SERVER, PLUS PINGS), 1-SWITCH MININET/RYU TESTBED, ∼99 S CAPTURE PER RUN, 10 RUNS PER MODE, MEAN±95% CI. ENTRIES = FLOW ENTRIES RETURNED PER FLOW-STATISTICS REPLY (1 S POLLING).

Mode	TCP PacketIns	All PacketIns	FlowMods	MAC-pair entries	IP-specific entries
MAC-pair (as ONOS in InSDN)	12.3±2.3	76.8±1.8	19.1±1.8	9.5±0.0	0.0±0.0
5-tuple, 10 s idle	10422.0±202.0	10497.0±204.7	10442.8±200.8	0.0±0.0	153.1±2.4
5-tuple, 2 s idle	10391.6±249.4	10611.3±258.7	10556.4±252.3	0.0±0.0	36.9±0.5

packets), but we did not isolate this mechanism. FlowMods (19.1 vs. about 10 440 and 10 560 per run) track the PacketIns and are not independent evidence. The MAC-pair result is a replication: the same mode in the first, flawed campaign gave 11.0 ± 4.1. Seeds 0–4 are the earlier five runs per mode (reported in the previous version of this study); seeds 5–9 were added later.

Idle timeout sets what a poller sees. Per-flow entries returned per statistics reply fell from 153.1 ± 2.4 to 36.9 ± 0.5 when the idle timeout dropped from 10 s to 2 s, a ratio of 0.234 to 0.246 in all ten pairs: entries expire between polls, so a 1 s poller sees about a quarter as many. The result is close to what a shorter timeout implies; its value is the size of the loss for a realistic polling interval. In MAC-pair mode the replies contain only the table-miss rule and MAC-pair entries (9.5 ± 0.03 per reply); no entry matches IP or transport fields, as in InSDN (whose ONOS additionally installed `eth_type` punt rules).

A flawed first campaign. Our first application installed MAC-pair entries for ping and ARP in every mode; these also match later TCP between the same hosts, so most connections never reached the controller and per-flow mode produced only 43.4 ± 30.1 TCP PacketIns per run (4× MAC-pair) with a wide interval. The original application is archived as `ryu_fwd_v1.py`. We report it because it shows how an overlapping coarse entry hides traffic from the controller by two orders of magnitude, independently of the forwarding mode's nominal granularity.

Reliability. One run (5-tuple, 2 s idle, seed 4) had 41 failed requests (7–10 per client); the other 29 one-switch runs and all 30 three-switch runs had none. The run did not reproduce, and its failures were spread evenly over all five clients, which is consistent with a short stall of the shared path or controller rather than one client. The cause is not established (the failure reasons were not logged in that run; later runs log them). The run is included in all figures and tables. Sensitivity analysis confirms that this run does not alter our substantive conclusions: excluding seed 4 shifts mean TCP PacketIns from $10\,391.6 \pm 249.4$ to $10\,464.3 \pm 213.6$ (+0.7%) and FlowMods from $10\,556.4 \pm 252.3$ to $10\,629.7 \pm 216.8$, while flow entries per reply remain virtually identical (36.8 ± 0.6 vs. 36.9 ± 0.5).

A second topology. On a three-switch chain (hosts h1–h6, server two hops from h1 and h2; 10 runs per mode, 26 826 requests, none failed) the contrast holds (Table XVIII): 18.0 ± 5.0 TCP PacketIns per run under MAC-pair against $10\,881 \pm 794$ and $10\,676 \pm 804$ under per-flow forwarding (391× to 1 509× in every paired run), and the per-flow entries a 1 s poller sees per reply fall from 112.4 ± 2.3 to 27.0 ± 0.4 (ratio 0.235 to 0.245). The ratio is smaller than on one switch because the MAC-pair baseline rises with the extra hops (18.0 vs. 12.3).

A stock Ryu application. To check that the MAC-pair behaviour is not an artefact of our own application, we ran Ryu's unmodified `simple_switch_13` on the same one-switch topology (5 runs of 60 s; 2 982 requests, none failed). Of 91 FlowMods, 86 match `(in_port, eth_dst, eth_src)` and 5 are the table-miss rule; none matches IP or transport fields. The controller received 71, 69, 76, 76 and 76 PacketIns per run (mean 73.6, all protocols). Like ONOS's `fwd`, this application installs MAC-pair entries; this follows from the application's code, and it does not test OpenDaylight, ONOS under other applications, or any production deployment.

Scope. This is a testbed with at most three switches with one forwarding application and benign traffic only; we run no attacks or detectors in it. The mode contrast does not transfer to InSDN's ONOS, which punted about one packet per target packet even with MAC-pair entries (§IV), and no claim is made about detection accuracy under any forwarding mode.

X. TARGETED LITERATURE AUDIT AND FEATURE-RECIPE EVALUATIONS

A. Targeted Audit of SDN-IDS Literature

To assess how prevalent these evaluation flaws are in the SDN-IDS literature, we conducted a targeted audit of 12 representative papers (eleven focusing on SDN or InSDN, and one non-SDN control). As documented in Table XII, ten papers were read in full text.

Our audit finds recurring methodological gaps:

1) **Absence of Host Grouping:** Among all InSDN papers read in full, none reports a host-grouped or capture-grouped validation split. Most employ simple randomized partitions (such as `train_test_split`) or fail to document their split mechanism entirely.
2) **Use of Network Identifiers as Model Inputs:** Three papers explicitly include non-generalizable network identifiers (Flow IDs, IP addresses, or port numbers) among their feature sets. Notably, in SDNGuardStack [5] the top-15 mutual-information features include Flow ID (ranked first), Src IP, Dst IP, Src Port and Dst Port, and the authors' SHAP ranking puts Flow ID first. OptiXGB-IDS [6] and Mehmood et al. [7] retain source or destination port as inputs; in the former the surrogate tree's major split is Dst Port, and in the latter Dst Port is the strongest negative SHAP contributor.
3) **Offline Evaluation of Controller Claims:** Multiple papers frame their models as lightweight controller-side defenses, yet evaluate exclusively on offline CSVs without deploying within an SDN controller.

TABLE XVIII
FORWARDING APPLICATION VS. WHAT THE CONTROLLER OBSERVES: BENIGN TRAFFIC FROM THE SAME GENERATOR AND SEEDS (5 HOSTS → 1 HTTP SERVER, PLUS PINGS), 3-SWITCH LINEAR MININET/RYU TESTBED, ∼99 S CAPTURE PER RUN, 10 RUNS PER MODE, MEAN±95% CI. ENTRIES = FLOW ENTRIES RETURNED PER FLOW-STATISTICS REPLY (1 S POLLING).

Mode	TCP PacketIns	All PacketIns	FlowMods	MAC-pair entries	IP-specific entries
MAC-pair (as ONOS in InSDN)	18.0±5.0	267.8±18.2	33.9±3.8	7.0±0.0	0.0±0.0
5-tuple, 10 s idle	10881.3±793.8	11159.9±795.4	10928.3±791.8	0.0±0.0	112.4±2.3
5-tuple, 2 s idle	10675.5±803.9	11275.8±814.7	11040.4±805.9	0.0±0.0	27.0±0.4

B. Feature-Recipe Evaluations on InSDN

To determine whether the evaluation gap is specific to feature selection, we implemented four generic feature recipes approximating choices in the literature, evaluated with GPU XGBoost across 10 seeds (Table XI):

1) **Numeric features only (77 features):** Random split achieves 0.9998 BAcc (0.9991 8-class); source-IP-disjoint evaluation drops to 0.943 BAcc (0.788 8-class).
2) **Numeric + ports (79 features):** Random split achieves 0.9998 BAcc (0.9998 8-class); source-IP-disjoint evaluation drops to 0.856 BAcc (0.687 8-class).
3) **Numeric + identifiers (83 features):** Random split achieves 1.0000 BAcc (0.9999 8-class); source-IP-disjoint evaluation drops to 0.941 BAcc (0.804 8-class).
4) **Top-15 by Mutual Information:** Selected on full data prior to splitting. Mutual information automatically selects Flow ID, Src IP and Dst IP. Random split achieves 1.0000 BAcc (0.9999 8-class); source-IP-disjoint evaluation drops to 0.941 BAcc (0.835 8-class, widest CI 0.241).

Across all recipes, random splits achieve near-perfect scores (≥ 0.9998), but performance degrades under source-IP-grouped evaluation. We note that Table XI evaluates an 80:20 split (standard in published literature), where 80% of source addresses are retained in training, raising hypergeometric primary-host retention to $0.80^2 = 64\%$. This explains why numeric features score 0.943 under 80:20 grouping compared to 0.907 under 70:30 grouping (Table X).

XI. METHODOLOGICAL GUIDELINES AND EVALUATION CHECKLIST

To improve rigor in SDN intrusion detection research, we propose seven concrete methodological guidelines:

1) **Verify Controller Observability:** Researchers proposing controller-side defenses must verify that input features correspond to OpenFlow control-plane primitives. If features require full-packet timing moments, the architectural cost of data-plane polling or flow-table modifications must be explicitly accounted for.
2) **Report Address-Disjoint Splits and Multi-Protocol Partitions:** Random flow splits should not be the sole basis for claims of generalizable detection. Report address-disjoint evaluations alongside (source-IP, destination-IP, and pair-disjoint), and explicitly state how many dominant endpoints each partition isolates.
3) **Isolate Address Spoofing from Host Identity:** In datasets containing volumetric flooding, researchers must distinguish ephemeral spoofed addresses from stable physical endpoints. Grouping by source IP alone does not evaluate unseen-host generalization when attack tools randomize the source address.
4) **Purge Non-Generalizable Network Identifiers:** All non-statistical network metadata (such as Flow IDs, IP addresses, MAC addresses, and ephemeral transport ports) must be removed prior to feature extraction.
5) **Audit Host Concentration and Apply Variance-Moderated Weighting:** Authors must report the distribution of flows per host. In datasets where a handful of hosts dominate total volume, Leave-One-Source-IP-Group-Out (LOSGO) testing should be reported alongside aggregate metrics, and researchers should employ square-root frequency attenuation ($w_i \propto 1/\sqrt{N_{\text{host}}}$) to prevent dominant hosts from monopolizing gradient representation without destabilizing training.
6) **Evaluate Forwarding Application Sensitivity:** Detectors relying on control-plane metrics (such as PacketIn rates) must evaluate sensitivity to forwarding application policies, entry granularity, and idle timeout configurations.
7) **Report Session-Identity Baselines and Label Coverage:** When benign and attack traffic are recorded in separate sessions, report trivial identity rules (an any-IP rule and a host-memory rule learned on training data), the attack windows per deployment and class, and, where possible, record benign and attack traffic interleaved.

XII. DISCUSSION AND LIMITATIONS

A. Scope and Experimental Boundaries

The controller-side results concern one benchmark (InSDN), two controllers (ONOS with default reactive forwarding and Ryu with `simple_switch_13`) and two deployments (Metasploitable-2 and OVS). CICIDS2017 is a non-SDN reference used only for the split analysis. We attempted a third candidate, UKMDDoSDN [42], but its public PCAPs contain no benign traffic and no OpenFlow control channel (Appendix A). We frame this work as a measurement study and methodological critique, not as a deployable end-to-end detector.

B. Threats to Validity

Construct validity. Window labels come from matching PacketIn 5-tuples to CSV records that were produced from separate end-host recordings; the alignment of the two recordings is unverified. Labelled attack windows are those in which

a CSV-matched flow crossed the controller, which may favour easy windows, and OVS DDoS and OVS brute-force have no labelled attack windows (Table II). Address-disjoint splits enforce IP separation, which cannot be equated with physical host isolation under address spoofing.

Internal validity. Seed intervals reflect initialization and fold assignment, not the sampling of captures (the capture-level bootstrap addresses the latter). The address-disjoint results rest on correlated splits in which eight hosts hold 62.6% of the rows, so the address logging shows an association, and leave-one-source-IP-group-out is a single case per address group. Our flow reassembler follows CICFlowMeter's default expiration rules but is not CICFlowMeter, and part of the mismatch with the CSV may stem from that difference. The host-identity rules are simple, so their score is a lower bound on what identity can explain.

External validity. ONOS and Ryu reactive forwarding were studied. Proactive rule installation, hardware TCAM limits, multiple controllers, and production traffic are outside the scope.

XIII. Conclusion

We audited what an SDN controller observes in InSDN and how that benchmark is evaluated. In the 44 ONOS captures, no flow-table entry or FlowMod matches IP addresses or ports, so per-flow counters are absent, while an IPv4 punt rule delivers packet headers to the controller; software reassembly of flows from those PacketIns is partial (38% packet correspondence on DoS in the examined unsynchronized recordings, $\rho = -0.27$ duration correlation). A controller-side MLP reaches 0.946 balanced accuracy on unseen captures, but because benign and attack traffic were recorded in separate sessions, trivial rules based on IP presence or on host identity explain much of the separation, and the score cannot be read as attack detection. Under strict labelling, OVS DDoS and OVS brute-force carry no attack windows and probing dominates the rest.

On flow features, random splits give ≥ 0.9917 balanced accuracy on both InSDN and CICIDS2017, whereas source-IP-disjoint evaluation gives 0.907 ± 0.085 and 0.641 ± 0.056. On genuine non-spoofed traffic, performance collapses to 0.8417 ± 0.1114 (Macro-F1 0.6503 ± 0.2005). The InSDN dispersion is consistent with which of eight dominant hosts is withheld (derived via exact hypergeometric host allocation and verified across 50 seeds, $r = -0.7713, p = 5.61 \times 10^{-11}$). Square-root frequency attenuation provides a constructive remedy raising mean balanced accuracy to 0.931 ± 0.064 (worst split 0.660 to 0.730), while pure inverse-frequency collapses due to 122,000 single-packet spoofed addresses. Leave-one-source-IP-group-out (LOSGO) recall on the primary attacker is 51.6% (DoS 34.7%). None of the 12 audited SDN-IDS papers reports an address-disjoint split. These results come from one dataset and one controller, and the split analysis confirms known leakage mechanisms on an SDN benchmark rather than introducing a new one. We offer seven evaluation guidelines and release the code and result files so that the measurements can be repeated and challenged.

Appendix A
Candidate Second Datasets Examined

In addition to InSDN and CICIDS2017, we examined candidate datasets for control-channel analysis. For UK-MDDoSDN [42], we fetched `syn_flood_combined.pcap` (18 922 240 B; 199 273 packets) and `udp_flood_combined.pcap` (15 420 748 B; 110 518 packets). Parsing confirmed that neither capture contains packets on TCP port 6633 or 6653; they represent data-plane host captures without OpenFlow control traffic, and contain no benign class. For LAN-SDN-NIDS [23], its record describes flow-level records with 11 control-plane metrics, but raw OpenFlow control-plane captures are not available. For ASEADOS-SDN-IoT [24], it provides 83 flow-level CSV features from an ONOS/IoT testbed. None of these datasets provided raw OpenFlow control captures to support control-channel reassembly.

Data and Code Availability

InSDN CSVs and PCAPs are publicly available (sources and SHA-256 digests in `data/MANIFEST.md`). The OpenFlow 1.3 parser, feature extraction, evaluation code, the per-experiment result files and the table and figure generators are in the project repository (https://github.com/adhyan-jain/Proteus, directory `research/insdn_controller`). The script `reproduce.sh` is provided to regenerate the result files from the raw data (it requires a CUDA GPU), and `paper/verify_numbers.py` checks every number quoted in the text against the stored result files. Experiments ran in Python 3.12 with PyTorch 2.11 (CUDA 12.8) and XGBoost 3.4 on one NVIDIA RTX 4060 laptop GPU. Capture hashes are in `capture_hashes.sha256`.

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